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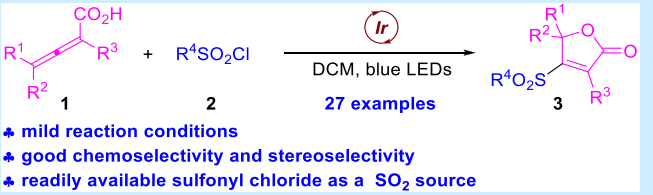
# Photocatalytic Chemoselective Cyclic Oxysulfonylation of 2,3-Allenoic Acids

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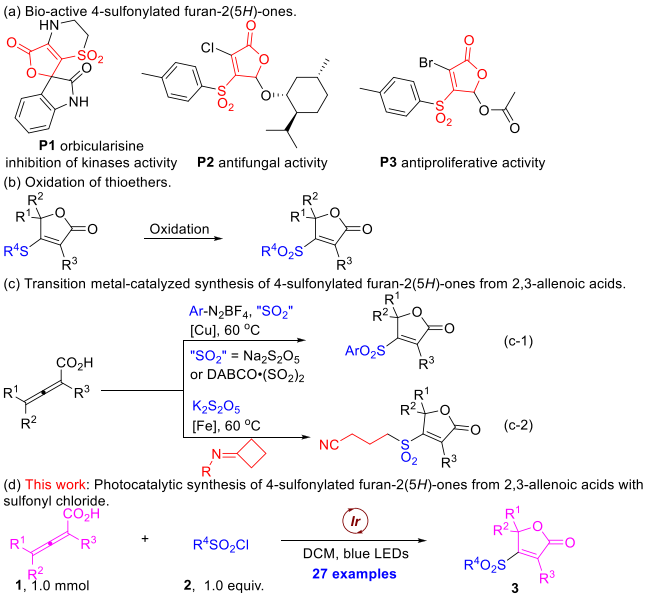
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**ABSTRACT:** Photoinduced cyclization of 2,3-allenoic acids with sulfonyl chloride providing an efficient synthesis of 4-sulfonylated furan-2(5*H*)-ones under mild reaction conditions has been achieved. The reaction enjoys a high chemoselectivity and tolerates a wide range of functional groups. The catalytic cycle has been validated through control experiments, cyclic voltammetry studies, and Stern−Volmer quenching studies.




4-Sulfonylated furan-2(5*H*)-ones are widely found in natural products and bioactive molecules (Scheme 1a).[1,2] Therefore,

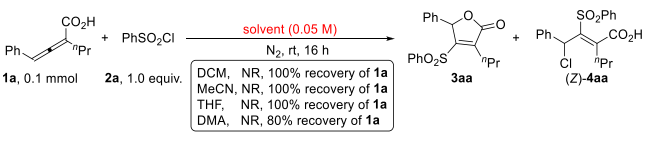
**Scheme 1. Typical Examples of Bioactive 4-Sulfonylated Furan-2(5*H*)-ones and Synthesis of 4-Sulfonylated Furan-2(5*H*)-ones from 2,3-Allenoic Acids**




methods to access this sort of skeleton are of great significance. The traditional pathway to afford these compounds is the oxidation of 2(5*H*)-furanonyl thioethers in the presence of stoichiometric amounts of strong oxidants (Scheme 1b).[3] On the other hand, allenoic acids have been widely employed to synthesize furan-2(5*H*)-one derivatives due to their high reactivity.[4] In 2019, Wu and co-workers disclosed the $Cu(OAc)_2$-catalyzed cyclization of 2,3-allenoic acids with aryldiazonium tetrafluoroborates with 1.5 equiv of $Na_2S_2O_5$ or 1.0 equiv of DABCO·$(SO_2)_2$ as the $SO_2$ source at 60 °C (Scheme 1c-1).[5] Recently, this transformation was also realized with cyclobutanone oxime ester under $FeCl_2$ catalysis with 2.0 equiv of $K_2S_2O_5$ as the $SO_2$ source at 60 °C (Scheme 1c-2).[6] The direct sulfonylation of 2,3-allenoic acids with readily available sulfonyl chloride[7] should be an atom economic strategy, however, has never been realized. The challenge is how to efficiently break the S($O_2$)−Cl bond, control the regioselectivity referring to the two C═C bonds in allene, and control the stereoselectivity required for the cyclization. Herein, we realize such a reaction via a photocatalytic strategy at room temperature affording 4-sulfonylated furan-2(5*H*)-ones (Scheme 1d).

We began our studies with the coupling reaction of 4-phenyl-2-propyl-2,3-butadienoic acid **1a** and benzenesulfonyl chloride **2a** (2.0 equiv). First of all, the cyclization was tried under the traditional electrophilic cyclization conditions (Scheme 2);[8] however, the expected cyclization product, 4-sulfonylated furan-2(5*H*)-one **3aa**, was not observed with the starting materials untouched.

**Scheme 2. Cyclization Reaction of 1a with 2a under Traditional Electrophilic Cyclization Conditions**




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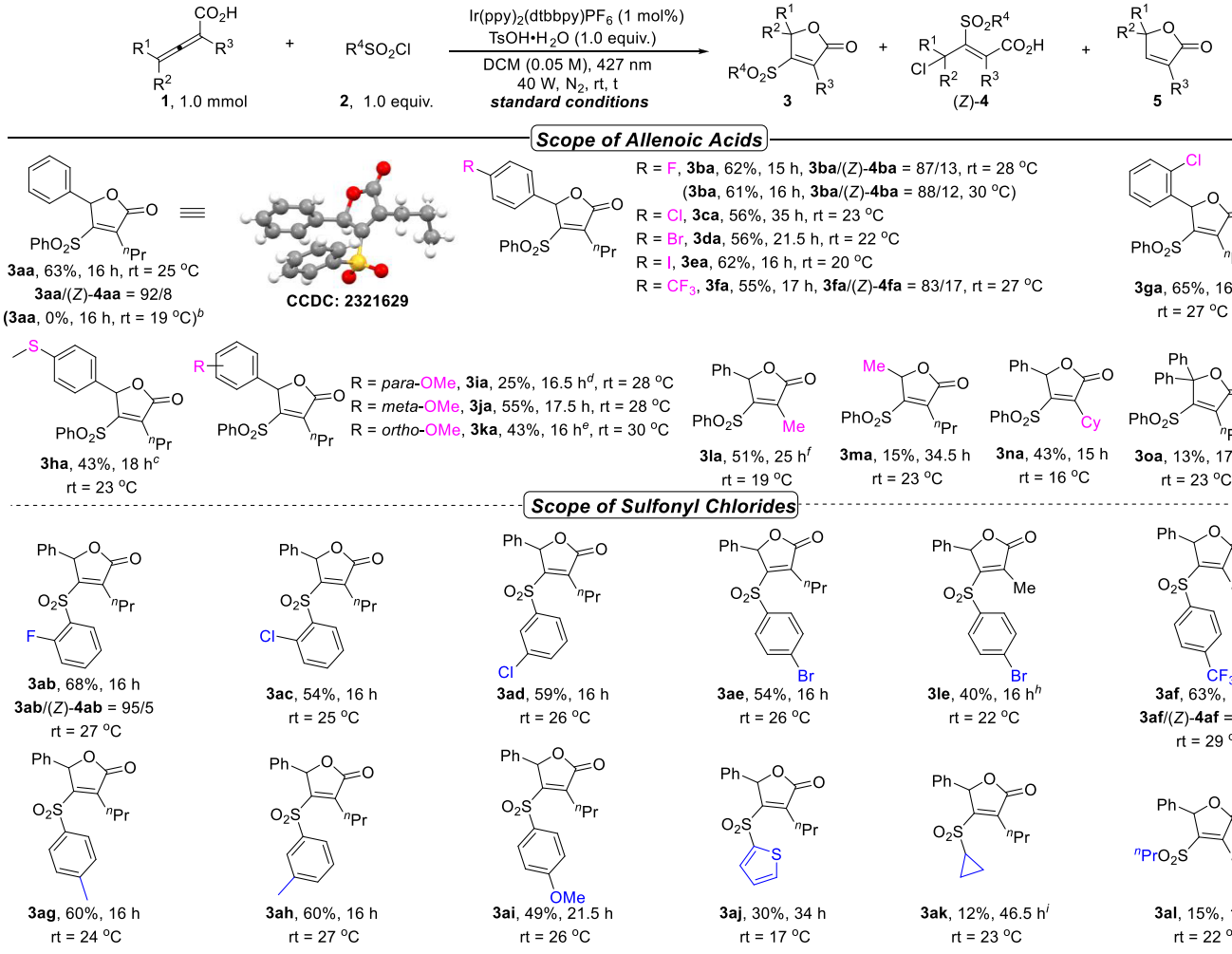
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**Table 1. Substrate Scope**[a]




[a]Reaction conditions: allenoic acids **1** (1.0 mmol), sulfonyl chlorides **2** (1.0 mmol), $TsOH{\cdot}H_2O$ (1.0 mmol), and $Ir(ppy)_2(dtbbpy)PF_6$ (0.01 mmol) in DCM (20 mL) at room temperature under $N_2$ atmosphere. The distance between the reaction tube and the light source is 4−6 cm, unless otherwise noted. [b]The reaction was conducted under air atmosphere on 0.1 mmol scale. The recovery of **1a** was 7% by NMR. [c]46% of **5h** was obtained by isolation. [d]52% of **5i** was obtained by isolation. [e]28% of **5k** was obtained by isolation. [f]5% of **5l** was obtained by NMR. [g]75% of **5o** was obtained by isolation. [h]4% of **5l** was obtained by NMR. [i]6% of **5a** was obtained by isolation. [j]16% of **5a** was obtained by isolation. The ratio of **3aa**/(*Z*)-**4aa** was determined by isolation while the ratio of **3ba**/(*Z*)-**4ba**, **3fa**/(*Z*)-**4fa**, **3ab**/(*Z*)-**4ab**, and **3af**/(*Z*)-**4af** were determined by $^{19}F$ NMR.

Interestingly, under visible-light irradiation, the desired product, 5-phenyl-4-(phenylsulfonyl)-3-propylfuran-2(5*H*)-one (**3aa**) was formed in 58% with 10% chlorosulfonylation product (*Z*)-**4aa** as a byproduct when 4CzIPN was employed as the photocatalyst (Table S1, entry 1; for details, see the Supporting Information), indicating that the stereoselectivity is indeed an issue. After testing a series of photocatalysts, $Ir(ppy)_2(dtbbpy)PF_6$ was found to offer **3aa** in an optimal yield of 76% with 16% chlorosulfonylation product (*Z*)-**4aa** (Table S1, entries 2−5). Solvent screening indicates that DCM was still the best (Table S1, entries 6−8). The reaction may be conducted with 1.0 equiv of **2a** with a slightly higher chemoselectivity (Table S1, entry 5 vs entry 9). Furthermore, it was found that, when the reaction was conducted with a concentration of 0.03 M, the yield of **3aa** was 82% with 9% yield of (*Z*)-**4aa** (Table S1, entry 10). Subsequently, we also screened some additives such as $TsOH{\cdot}H_2O$, AcOH, and NaOAc (Table S1, entries 12−14), and $TsOH{\cdot}H_2O$ delivered the best result. Control experiments revealed that both the light and the photocatalyst were essential for the transformation (Table S1, entries 15 and 16). Running the reaction on 1.0 mmol scale at 0.05 M has been defined as the optimal conditions, affording **3aa** in 69% yield with a remarkable selectivity of **3aa**/(*Z*)-**4aa** (Table S1, entry 18). Furthermore, the reaction with 427-nm blue LEDs is still the optimal choice after the light source test (Table S1, entries 19−21). The structure of byproduct (*Z*)-**4aa** was further confirmed by (*Z*)-**4le** being esterificated to afford (*Z*)-**10le**, which has been clearly identified by X-ray diffraction (XRD) analysis (CCDC No. 2127599; for details, see page S35 in the Supporting Information).

The scope of 2,3-allenoic acids and sulfonyl chlorides has been explored, as shown in Table 1. The cyclization reaction of 2,3-allenoic acid **1a** with benzenesulfonyl chloride **2a** afforded the corresponding product **3aa** in 63% yield with a selectivity of 92/8 for **3aa**/(*Z*)-**4aa**. When the reaction was conducted in air, the target product **3aa** was not observed, with 7% recovery of **1a**. The structure of **3aa** was further confirmed by X-ray



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analysis. Different electron-withdrawing-group-substituted 4-aryl 2,3-allenoic acids **1b**−**1g** performed well under the standard conditions, affording the corresponding products **3ba**−**3ga** in good yields. When 2,3-allenoic acid with the methylthio-substituted aryl moiety **1h** was applied, the desired product **3ha** was obtained in 43% yield. Moreover, 2,3-allenoic acids bearing a methoxy group at the *para*-, *meta*-, and *ortho*-position of the 4-aryl group all afforded the desired products **3ia**−**3ka** in moderate yields. It is worth mentioning that 46%, 52%, and 28% of cycloisomerization products **5h**, **5i**, and **5k** were generated for *para*-SMe **1h**, *para*-OMe **1i**, and *ortho*-OMe **1k**, respectively.

Furthermore, the *n*-propyl group in **1a** could be replaced with a methyl group to provide **3la** in 51% yield. However, 4-methyl-substituted 2,3-allenoic acid **1m** afforded the desired product **3ma** in a low yield of 15%. When $R^2 = 2°$ alkyl group, i.e., 2,3-allenoic acid **1n**, the reaction could also afford desired product **3na** in 43% yield. It should be noted that tetrasubstituted 2,3-allenoic acid **1o** afforded the desired product **3oa** in 13% yield with 75% cycloisomerization product **5o**. With regard to sulfonyl chlorides, the substrates tethered with various synthetically versatile functional groups, such as F, Cl, Br, $CF_3$, Me, and OMe, all afford the corresponding products **3ab**−**3ai** in moderate yields. Thienyl sulfonyl chloride **2j** also delivered product **3aj** in 30% yield. However, the cyclopropylsulfonyl chloride **2k** afforded the corresponding product **3ak** in a low yield of 12% with 6% cycloisomerization product **5a**. And *n*-propylsulfonyl chloride **2l** afforded the desired product **3al** in only 15% yield with 16% of cycloisomerization product **5a**. The reactions of **1a** with 1-naphthalenesulfonyl chloride **2m**, dimethylamino sulfonyl chloride **2n**, or D(+)-10-camphorsulfonyl chloride **2o** all failed to afford **3am**, **3an**, or **3ao** under the standard conditions (for details, see pages S42−S44 in the Supporting Information).

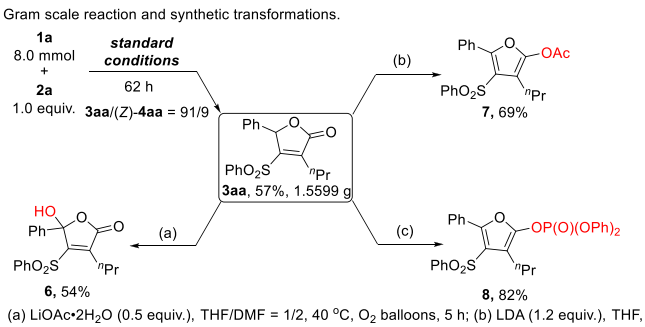
A gram-scale reaction was conducted under the standard reaction conditions, affording **3aa** in 57% yield with a selectivity of 91/9 for **3aa**/(*Z*)-**4aa**. In the presence of $LiOAc{\cdot}2H_2O$, the *γ*-position of **3aa** could be easily oxidized to give *γ*-hydroxylated product **6** in 54% yield under an $O_2$ atmosphere. Deprotonation and subsequent acetylation with $Ac_2O$ afforded sulfonylated furan **7**. Similarly, an O−P bond could be formed when **3aa** was treated with NaHMDS and subsequent phosphorylation with $ClP(O)(OPh)_2$ to offer **8** in 82% yield (Scheme 3).[8b,9]

To gain insight into the mechanism, control experiments were conducted. First, when the reaction of **1a** with $PhSO_2Cl$ was performed in the presence of 1,1-diphenylethylene (2.0 equiv) under the standard conditions, the formation of **3aa** was totally suppressed and the sulfonylation product of 1,1-diphenyl-2-benzenesulfonylethylene **9** was produced, indicating the presence of sulfonyl radical in this transformation (Scheme 4a). When the byproduct chlorosulfonylation product (*Z*)-**4aa** was submitted to the standard conditions, desired product **3aa** was also observed in 19% yield with 58% recovery of (*Z*)-**4aa** (Scheme 4b). Cyclic voltammetry experiments confirmed that both 4-phenyl-2,3-allenoic acid **1a** ($E^{red}_{1/2}$ [**1a**/**1a**$^{\bullet -}$] = −0.81 V vs SCE in DCM) and benzenesulfonyl chloride **2a** ($E^{red}_{1/2}$ [**2a**/**2a**$^{\bullet -}$] = −1.06 V vs SCE in DCM) could be reduced by the excited photocatalyst $*Ir(ppy)_2(dtbbpy)PF_6$ ($E^{red}_{1/2}$ [$Ir^{IV}/Ir^{III}*$] = −1.30 V vs SCE in DCM). It was found that **2a** more easily underwent a single-electron transfer (SET) process with the excited photocatalyst. As expected, 4-methyl-substituted 2,3-allenoic acid **1m** ($E^{red}_{1/2}$ [**1m**/**1m**$^{\bullet -}$] < −2.00 V vs SCE in DCM), *n*-propylsulfonyl chloride **2l** ($E^{red}_{1/2}$ [**2l**/**2l**$^{\bullet -}$] = −1.39 V vs SCE in DCM), and cyclopropylsulfonyl chloride **2k** ($E^{red}_{1/2}$ [**2k**/**2k**$^{\bullet -}$] = −1.48 V vs SCE in DCM) showed higher reduction potentials, indicating **1m**, **2l**, and **2k** were difficult to be reduced (Scheme 4c). However, Stern−Volmer quenching studies showed that **2a** or **2a**/$TsOH{\cdot}H_2O$ could quench excited photocatalyst $*Ir(ppy)_2(dtbbpy)PF_6$ more efficiently than **1a**. This result indicated that **2a** would be the exact quencher of the excited photocatalyst in the reaction. Furthermore, **2l**, **2k**, and **2j** were all demonstrated to display a lower quenching efficiency with the excited photocatalyst, compared with that of **2a** (Scheme 4d).

Based on these experimental data, a possible mechanism is depicted below using **1a** and **2a** as the model substrates (Scheme 5). The excited photocatalyst $*[Ir]^{III}$ could be quenched with **2a** to generate $[Ir]^{IV}$ and a benzenesulfonyl radical, which would react with allenoic acid **1a** to form radical *anti*-**Int I** as the major intermediate and its stereoisomer *syn*-**Int I** as the minor intermediate. Subsequent oxidation of radical intermediates, *anti*-**Int I** and *syn*-**Int I**, with $[Ir]^{IV}$ species would afford ground-state photocatalyst $[Ir]^{III}$ and cationic allylic intermediates, *anti*-**Int II** or *syn*-**Int II**, via a SET process. **3aa** was generated from *anti*-**Int II** by intramolecular nucleophilic substitution. Meanwhile, *syn*-**Int II** could be trapped with $Cl^-$ to form noncyclized chlorosulfonylation product (*Z*)-**4aa**. When the phenyl group in **1a** was replaced with a methyl group, there is no benzyl effect; thus, the reaction of **1m** gave a poor yield of **3ma** (Table 1). The reactions of **1a** with **2j**, **2k**, and **2l** afforded the desired products **3aj**, **3ak**, and **3al** in poor yields (Table 1), probably because these compounds quenched the excited photocatalyst in a mismatched rate (Scheme 4d). It should be noted that (*Z*)-**4aa** could be converted to the target molecule **3aa** at a slow rate.

In summary, a photocatalyzed cyclic oxysulfonylation reaction of 2,3-allenoic acids with sulfonyl chlorides has been developed. This protocol enables to prepare the sulfonylated furan-2(5*H*)-ones with a high chemoselectivity and a decent functional group tolerance under mild conditions. The synthetic application has been proven via versatile transformations of the target molecules. Radical trapping experiments and cyclic voltammetry experiments unveiled that sulfonyl radical generated via light-induced SET process were involved via the reductive quenching cycle. In addition, cyclic voltammetry experiments and Stern−Volmer quenching

**Scheme 3. Synthesis of 3aa on a Gram Scale and Derivatizations**




(a) $LiOAc{\cdot}2H_2O$ (0.5 equiv.), THF/DMF = 1/2, 40 °C, $O_2$ balloons, 5 h; (b) LDA (1.2 equiv.), THF, -78 °C, 0.5 h, then $Ac_2O$ (2.0 equiv.), -78 °C, $N_2$, 4 h; (c) NaHMDS (1.2 equiv.), THF, -78 °C, $N_2$, 0.5 h, then $ClP(O)(OPh)_2$ (2.0 equiv.), -78 °C, $N_2$, 4 h.

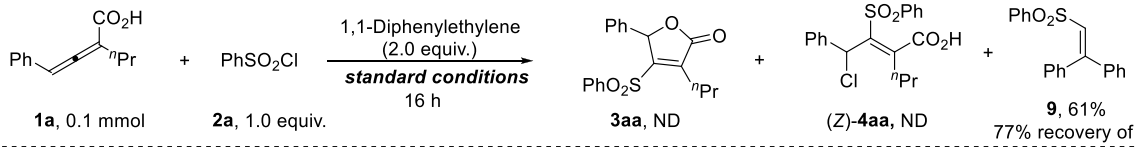
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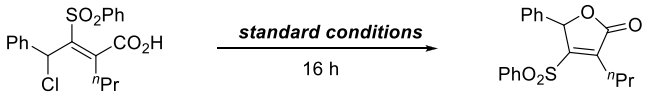
**Scheme 4. Mechanistic Experiments**

(a) Radical trapping experiment.



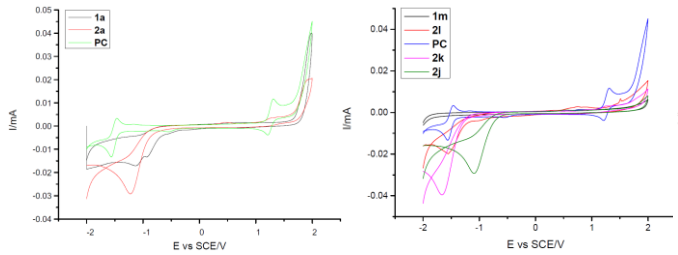

**1a**, 0.1 mmol **2a**, 1.0 equiv. **3aa**, ND (*Z*)-**4aa**, ND **9**, 61% 77% recovery of **1a**

(b) Control experiments for the photocatalysed cyclization of (*Z*)-**4aa**.

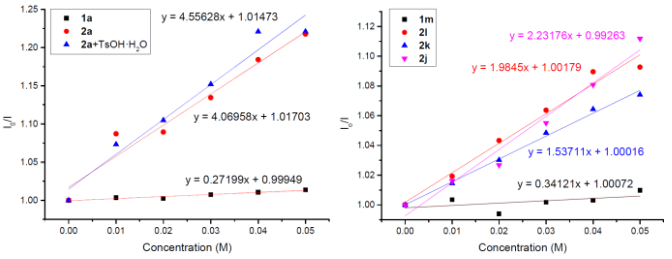



(*Z*)-**4aa** 0.12 mmol **3aa**, 19% NMR yield 58% recovery of (*Z*)-**4aa**

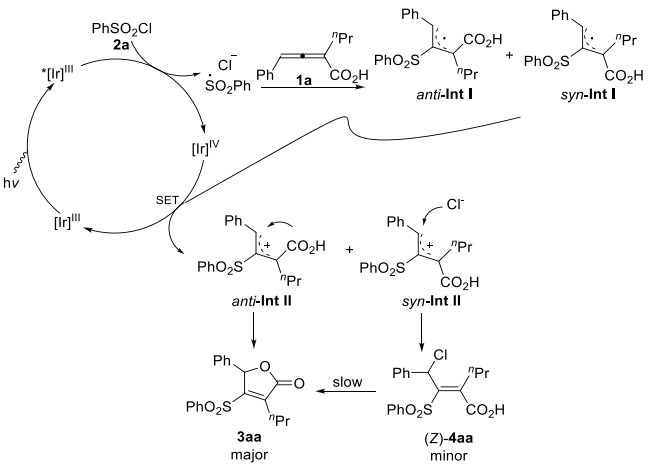
(c) Cyclic voltammetry experiments. (The starting point of CV plot was -2 V and scan direction was from -2 V to +2 V.)




(d) Stern-Volmer quenching experiments.




**Scheme 5. Proposed Mechanism**




studies indicated that the aryl sulfonyl chlorides perform much better than the alkyl sulfonyl chlorides. Further studies are being pursued in our laboratory.

## ■ ASSOCIATED CONTENT

### Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c01730.

Experimental procedures, compound characterization data (PDF)

### Accession Codes

CCDC 2127599 and 2321629 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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### Notes

The authors declare no competing financial interest.

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Zhejiang University. We also thank Mr. Dong Huang in our group for reproducing the results of **3ka**, **3ab**, and **3ai**.

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